

Employment

Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Shipped 10+ production TypeScript/Node.js AI features: multi-model agent pipelines (Claude/GPT-4o/Gemini, MCP), Temporal.io durable workflows, real-time event delivery, and Auth0 RBAC; 923 commits, 7 deployments. - Own backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry); maintain CI/CD and resolve production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deploy genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. - Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. - Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

AlgoPulse	Agentic AI Orchestration Platform	2025
- Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load. - Replaced API polling with Supabase Edge Functions for real-time AI streaming to clients; eliminated state loss under peak traffic.		
VectorLift	Semantic Search & RAG Evaluation Pipeline	2026
- Built production semantic search on MS MARCO: Elasticsearch BM25 baseline, sentence-transformers bi-encoder fine-tuned with in-batch negatives, MiniLM cross-encoder re-ranker; evaluated 6 pipelines with NDCG@10, MRR@10, MAP, and bootstrap tests. - Designed Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant vector store; served search via FastAPI with per-stage latency tracking and Streamlit experiment dashboard.		
LLM Eval Harness	Vendor-Neutral LLM Evaluation Framework	2025
- Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, multi-hop reasoning, and FEVER claim verification tasks. - Tracked hallucination rate, p50/p95 latency, and cost-per-call; applied bootstrap CIs and McNemar tests across 6 metrics per model.		

Languages and Technologies

- Languages:** Python (asyncio, FastAPI), TypeScript, JavaScript, Java, SQL
- AI/Agentic:** Multi-agent systems, RAG, vector databases (Qdrant, Elasticsearch), LLM evaluation, fine-tuning, MCP
- Backend & Infra:** Node.js, NestJS, Temporal.io, Kafka, PostgreSQL, Redis, MongoDB, Docker, Kubernetes
- Cloud & Observability:** AWS, GCP, Terraform, OpenTelemetry, Prometheus, Grafana, Sentry, GitHub Actions

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- Technical Writing:** Published on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog